Econometrics Midterm

Short Answer · Econometrics 1–3 (Slides), Wooldridge Ch.1–6, Quiz 2 · 37 questions

1. What is the difference between a population and a sample?

2. Why do econometricians usually study a sample instead of the population, and what is the goal of studying the sample?

3. Define external validity and state which property of the sample matters for it.

4. In $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$, why is β_1 called a partial slope rather than simply "the slope"?

5. In $\log(\text{price}) = \beta_0 + \beta_1 \text{battery} + \beta_2 \text{sunlight} + \beta_3 \text{battery} \times \text{sunlight} + u$ with $\beta_3 \neq 0$, why does battery not have a linear relationship with log(price)?

6. What is the difference between random sampling and random assignment?

7. For Model 2, $\text{wage} = \beta_0 + \beta_1 \text{educ} + u$ (ability not included), explain why the zero conditional mean condition may be violated.

8. What is a control variable, and why is ability added in Model 1, wage = $\beta_0 + \beta_1$ educ + β_2 abil + u?

9. What does "ceteris paribus" mean, and why is it important in econometrics?

10. What is panel (longitudinal) data, and what is one advantage of it?

11. What does OLS minimize, and what is a residual?

12. What does R-squared measure in a regression?

13. What does the homoskedasticity assumption (SLR.5) state?

14. What does it mean that the OLS estimator β_1 is unbiased?

15. When does omitting a relevant-looking variable x_2 NOT cause bias in the estimated coefficient on x_1?

16. What is the main consequence of multicollinearity (high but not perfect correlation among regressors) for OLS?

17. State the Gauss–Markov theorem.

18. Define the p-value of a hypothesis test.

19. Does a statistically significant coefficient always matter in practice? Answer in 1–2 sentences.

20. Interpret a 95% confidence interval for β_j.

21. Which can make the usual OLS t statistics invalid (no t distribution under H_0): (i) heteroskedasticity, (ii) a sample correlation of 0.95 between two regressors in the model, (iii) omitting an important explanatory variable? Explain briefly.

22. What does it mean that an estimator is consistent, and which assumptions are enough for OLS to be consistent?

23. pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, what is the sign of the inconsistency in $\tilde{\beta_1}$, and why?

24. Can cigs (average cigarettes smoked per day) have a normal distribution in the adult population? What does this imply for OLS inference?

25. How is the LM (n-R-squared) statistic for testing q exclusion restrictions computed, and what distribution does it follow?

26. If the dependent variable is divided by 1,000 (dollars → thousands of dollars), what happens to the OLS coefficients, standard errors, t statistics and R-squared?

27. What is a beta (standardized) coefficient, and why is it useful?

28. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions), give the turning-point formula for a quadratic and find where the marginal effect of sales becomes negative.

29. In log(wage) = $\beta_0 + \beta_1$ educ + β_2 educ· pareduc + β_3 exper + β_4 tenure + u (pareduc = both parents' education), what is the return to one more year of education, and what does β_2 measure?

30. Why is adjusted R-squared used, and when does adding a variable increase it?

31. Why is a 95% prediction interval for one individual outcome y^0 much wider than the 95% confidence interval for $E(y|x = x^0)$?

32. When the dependent variable is log(y), why is exp(predicted log(y)) not a good prediction of y, and how is it fixed?

33. Palm oil production is estimated as: production = 243.63 − 3.34 tax − 0.61 fuel − 0.03 fuel^2. Production is the amount of palm oil produced, tax is the tax rate (%), and fuel is the fuel price ($). What is the predicted amount of palm oil produced when the tax rate is 5% and the fuel price is $7? Write the equation with the numbers substituted (two decimal places).

34. The palm oil equation above is estimated with an intercept and three regressors (tax, fuel, fuel^2). What are the degrees of freedom when the sample size is 85? Write the formula and the numbers.

35. Worker productivity is estimated as: productivity = 3.36 + 0.06 log(training) + 2.53 log(experience) + 0.03 log(training) × log(experience). Productivity is shoes made per hour, training is training hours, experience is years of experience (log = natural log). What is the predicted number of shoes the worker makes in one hour with 2000 hours of training and 17 years of experience? Write the equation with the numbers substituted (two decimal places).

36. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of log(vaccination) when 1 reminder is sent and there are five ambassadors? Write the equation with the numbers substituted (two decimal places).

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Econometrics Midterm — Model answers

1. Econ 1
A population is the entire set of individuals, objects or data points we want to study, while a sample is a subset of the population that is selected for analysis.
Key results: entire · subset · selected

2. Econ 1
Studying the whole population is often too costly or impossible, so we study a sample
the goal is to find out what the population is like from the sample.
Key results: costly · impossible · population is like

3. Econ 1
External validity is the validity of applying the conclusions of a study outside the context of that study
it depends on whether the sample is representative of the population.
Key results: outside · conclusions · representative

4. Econ 2
β_1 is the change in log(price) from a one-unit increase in subsidy while holding the other variables (log(sunlight)) fixed, so it isolates the effect of subsidy alone.
Key results: holding · other variables · one-unit

5. Econ 2
Because battery and sunlight interact, the effect of battery on log(price) is $\beta_1 + \beta_3 \cdot$sunlight, which depends on the level of sunlight
the slope is not constant.
Key results: interact · depends · not constant

6. Econ 3
In random sampling every element of the population has an equal probability of being selected
in random assignment every element has an equal probability of being assigned to a group.
Key results: equal probability · selected · assigned

7. Econ 3
Ability is omitted, so it goes into the error u
because people with better ability get more education, educ is correlated with u, so $E(u|educ) = 0$ fails and β_1 is estimated with bias.
Key results: ability · error · correlated

8. Econ 3
A control variable is included to hold another factor fixed so it no longer sits in the error
adding ability helps satisfy the zero conditional mean condition and removes the bias in β_1.
Key results: hold · zero conditional mean · bias

9. Ch.1
Ceteris paribus means "other relevant factors being equal"
most economic questions ask for the causal effect of one variable holding all other factors fixed.
Key results: other factors · equal · causal

10. Ch.1
Panel data follow the same cross-sectional units over time
they can account for time-invariant unobservables and model lagged responses.
Key results: same · over time · time-invariant

11. Ch.2
OLS chooses the estimates that minimize the sum of squared residuals
a residual is the actual value y_i minus the fitted value $\hat{y}_i$.
Key results: sum of squared residuals · actual · fitted

12. Ch.2
R^2 = SSE/SST is the fraction of the sample variation in y that is explained by the explanatory variable(s).
Key results: fraction · variation · explained

13. Ch.2
Homoskedasticity means the error u has the same (constant) variance for any value of x: $Var(u|x) = \sigma^2$.
Key results: variance · error · same

14. Ch.2
Unbiasedness means $E(\hat{\beta}_1) = \beta_1$: if sampling and estimation were repeated many times, the estimates would on average equal the true population value, though one estimate may differ.
Key results: on average · repeated · true

15. Ch.3
There is no omitted variable bias if the omitted variable is irrelevant (its coefficient is zero) or uncorrelated with x_1.
Key results: irrelevant · uncorrelated

16. Ch.3
Multicollinearity increases the sampling variance of the affected slope estimators, but OLS remains unbiased
it can be detected with variance inflation factors.
Key results: variance · increase · unbiased

17. Ch.3
Under MLR.1–MLR.5, OLS is BLUE (best linear unbiased estimator): it has the smallest variance among all linear unbiased estimators.
Key results: BLUE · smallest variance · MLR.1

18. Ch.4
The p-value is the smallest significance level at which the null hypothesis would be rejected
we reject H_0 when the p-value is below the chosen level.
Key results: smallest · significance level · reject

19. Ch.4
Not necessarily: statistical significance only shows the effect differs from zero, while economic (practical) significance depends on the magnitude of the coefficient.
Key results: not necessarily · magnitude · economic

20. Ch.4
In repeated samples, intervals built this way contain the true population coefficient in 95% of cases
the bounds are random, not the parameter.
Key results: repeated · contain · true

21. Ch.4
Heteroskedasticity (i) and an omitted important variable (iii) violate the CLM assumptions, so the t statistics are invalid
a high correlation of 0.95 (ii) is not a CLM violation (only perfect collinearity is ruled out).
Key results: heteroskedasticity · omitted · correlation

22. Ch.5
An estimator is consistent if, as the sample size grows, its distribution collapses to the true parameter (plim $\hat{\beta}_j = \beta_j$)
OLS is consistent under MLR.1–MLR.4 (even zero correlation between each x_j and u is enough).
Key results: sample size · collapse · true · MLR.4

23. Ch.5
The inconsistency is positive: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1$, with $\beta_2 > 0$ (higher risktol raises pctstck) and $\delta_1 > 0$ because funds and risktol are positively correlated
it does not disappear as n grows.
Key results: positive · risktol · correlated · not disappear

24. Ch.5
No: most adults smoke zero cigarettes and cigs is nonnegative and discrete, so it is not normal
with a large sample, asymptotic normality still makes the t and F tests approximately valid.
Key results: zero · nonnegative · not normal · large sample

25. Ch.5
Estimate the restricted model, regress its residuals on all the explanatory variables, and compute LM = nR^2 from that auxiliary regression
under H_0 it is approximately chi-square with q degrees of freedom.
Key results: restricted · residuals · nR · chi-square

26. Ch.6
All coefficients and standard errors are divided by 1,000, while the t statistics and R-squared stay unchanged
only the units change, not the statistical conclusions.
Key results: divided · unchanged · t statistic · R-squared

27. Ch.6
A beta coefficient comes from regressing the standardized y on standardized x's
it is the change in y, in standard deviations, from a one standard deviation increase in x_j, so effects measured in different units can be compared.
Key results: standardized · standard deviation · units · compare

28. Ch.6
Turning point $\hat{x}^* = \hat{\beta}_1/(2|\hat{\beta}_2|) = 0.00030/(2 \times 0.0000000070) \approx 21{,}428.57$, i.e. sales of about $21.4 billion beyond that the estimated marginal effect of sales on rdintens is negative.

29. Ch.6
The return is $\beta_1 + \beta_2 \cdot$pareduc, so it depends on parents' education
the interaction coefficient β_2 is how much the return changes for one more year of pareduc (estimate 0.00078: at pareduc = 32 the return is about 7.2%).
Key results: depends · parents · interaction

30. Ch.6
R-squared never falls when a variable is added, so adjusted R-squared imposes a penalty for extra regressors using degrees of freedom
adding a variable raises it only if that variable's t statistic exceeds one in absolute value.
Key results: never falls · penalty · degrees of freedom · t statistic

31. Ch.6
The confidence interval reflects only the sampling variance of $\hat{y}^0$, while the prediction interval also includes the variance of the unobserved error for an individual outcome (σ^2), which does not shrink as n grows (e.g. colgpa: CI 2.66–2.74 vs PI 1.60–3.80).
Key results: error · variance · individual

32. Ch.6
$\exp(\log \hat{y})$ systematically underestimates $E(y|x)$
we multiply it by $\hat{\alpha}_0 = \exp(\sigma^2/2)$ under normal errors, or by the average of $\exp(\hat{u}_i)$ from the residuals.
Key results: underestimate · multiply · residuals

33. Ch.6
production $= 243.63 - 3.34(5) - 0.61(7) - 0.03(7)^2$
$= 243.63 - 16.70 - 4.27 - 1.47 = 221.19$
Key results: 221.19 · 16.7 · 4.27 · 1.47

34. Ch.3
df $= n - k - 1 = 85 - 3 - 1 = 81$
Key results: 81 · n − k − 1 · 85 − 3 − 1

35. Ch.6
ln(2000) $= 7.6009$ and $\ln(17) = 2.8332$
productivity $= 3.36 + 0.06(7.6009) + 2.53(2.8332) + 0.03(7.6009)(2.8332)$
$= 3.36 + 0.4561 + 7.1680 + 0.6460 = 11.63$
log base 10 would give 6.79, so log means ln
Key results: 11.63 · 7.60 · 2.83 · 7.17 · 0.65

36. Ch.6
log(vaccination) $= 0.34 + 0.15(1) + 0.23 \ln(5) = 0.34 + 0.15 + 0.23(1.6094) = 0.86$
Key results: 0.86 · 0.37 · 1.61

37. Ch.6
vaccination $= \exp(0.8602) = e^{0.8602} = 2.36$
Key results: 2.36 · 0.86